Syllabus Details

Syllabus ID:	14176
Syllabus Name:	**Software Requirement_Yêu cầu phần mềm**
Course Name English:	**Software Requirement**
Subject Code:	**SWR302**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 145-minute final exam + 102,6h self-study
Pre-Requisite:	SWE102 or SWE201c or SWE202c
Description:	This course is a model-based introduction to RE, providing the conceptual background and terminology on RE, addressing a variety of techniques for requirements development including Analysis and Requirements Elicitation; Requirements Evaluation; Requirements Specification and Documentation; Requirements Quality Assurance. To implement these frameworks, student will be learnt how to find appropriate customer representatives, elicit requirements from them, and document user requirements, business rules, functional requirements, data requirements, and nonfunctional requirements. The numerous visual models that will be represented to illustrate the requirements from various perspectives to supplement natural-language text. Other contents recommend the most effective requirements approaches for various specific classes of projects: agile projects developing products of any type, enhancement and replacement projects, projects that incorporate packaged solutions, outsourced projects, business process automation projects, business analytics projects, and embedded and other real-time systems.
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises given by instructor in class or at home and submit on time - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	- Microsoft office for documents and presentation. - https://www.visual-paradigm.com/download/community.jsp students must install visual-paradigm for drawing UML (offline) before taking Final Practical Exam - BOUML (http://bouml.free.fr/) : UML Drawing software
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	377/QĐ-ĐHFPT dated 04/09/2026
IsApproved:	**True**
Note:	In the case: (5 > Final TE Score >=4) & (5 > Final PE Score >=4) & FR < 5, the student can choose to take the resit of both TE & PE OR just either TE or PE.
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/9/2026

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Software Requirements	Karl E. Wiegers	Microsoft Press		Third Edition	978-0-7356-7966-5	True	True	False	
2	More About Software Requirements: Thorny Issues and Practical Advice	Karl E. Wiegers	Microsoft Press			978-0-7356-2267-8	False	True	False	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
3	The Software Requirements Memory Jogger: A Pocket Guide to Help Software And Business Teams Develop and Manage Requirements	Ellen Gottesdiener	GOAL/QPC			978-1-57681-060-6	False	True	False	
4	Requirements Engineering: Secure Software Specifications Specialization		Coursera				False	False	True	https://www.coursera.org/specializations/requirer engineering-secure-software

9 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Develop a good understanding of principles and techniques for requirement enginneering (RE) regarding requirement inception, requirement development(elicitation, analysis, specification, validation) and requirement management; understand main characteristics of specific projects
2	CLO2	Identify the appropriate requirements elicitiation techniques to identify requirements
3	CLO3	Utilize various requirements validation techniques to critically evaluate their requirements to identify defects
4	CLO4	Analysis of system requirements and the production of system specifications
5	CLO5	Create models of requirements using a variety of notations and techniques, including domain and usage models
6	CLO6	Design and plan software solutions to problems using an object-oriented strategy
7	CLO7	Prepare and deliver coherent and structured verbal and written technical reports
8	CLO8	Understand how to reduce risks by prototyping.
9	CLO9	Utilize the tactic of requirements management regarding changing, tracing and improving requirements

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Introduction The essential of software requirement - Software requirements defined	Offline	LO1	I	Chapter 1 + Slides	SWR302		
2	The essential of software requirement (cnt) - Requirements development and management - When bad requirements happen to good people - Benefits from a high-quality requirements process	Offline	LO1	I	Chapter 1 + Slides	SWR302		

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
3	Requirements from the customer's perspective - Who is the customer? - The customer-development partnership - Creating a culture that respects requirement	Offline	LO1	I,T	Chapter 2 + Slides			
4	Requirements from the customer's perspective (cnt) - Identifying decision makers - Reaching agreement on requirements Assignment introduction	Offline	LO1	I,T	Chapter 2 + Slides		do homework	
5	Good practices for requirements engineering (introduce) - A requirements development process framework - Requirements elicitation - Requirements analysis - Requirements specification - Requirements validation - Requirements management	Offline	LO1, LO2	I,T	Chapter 3 +Slides			
6	Good practices for requirements engineering (practice/apply)	Offline	LO1, LO2	I,T	Chapter 3 +Slides		do homework	
7	Group discussion Summary and Exercises (groups)	Offline	LO2, LO4, LO5		Chapter 4+ Business Analyst Job Description.docx + Slides		Discuss what students have to do in their Assignment	
8	Group discussion Summary and Exercises (groups)	Offline	LO2, LO4, LO5		Chapter 4+ Business Analyst Job Description.docx + Slides		Discuss what students have to do in their Assignment	
9	The business analyst - The business analyst role - The business analyst's tasks - Essential analyst skills - Essential analyst knowledge - The making of a business analyst - The analyst role on agile projects	Offline	LO2, LO4, LO5	I,T,U	Chapter 4+ Business Analyst Job Description.docx + Slides		Write Vision and Scope for Assignment	
10	Establishing the business requirements - Defining business requirements - Vision and scope document - Scope representation techniques	Offline	LO2, LO4, LO5	I,T,U	Chapter 5 + Vision and Scope Template.docx + Slides		Write Vision and Scope for Assignment	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
11	Establishing the business requirements (con't) - Keeping the scope in focus - Vision and scope on agile projects - Introduce LAB requirements: draw 1 context diagram, 1 swimlane diagram and 1 state diagram.	Offline	LO2, LO4, LO5		Chapter 5 + Vision and Scope Template.docx + Slides		Practice on first half of Assignment	
12	Finding the voice of users - User classes - User personal - Connecting with user representatives - The product champion - User representation on agile project - Resolving conflicting requirements	Offline	LO2, LO4, LO5		Chapter 6 + Product Champion Responsibilities.docx + Slides		Practice on first half of Assignment	
13	Requirements elicitation - Requirements elicitation techniques: Interviews, workshops, Focus groups, Observations, Questionaires, System interface analysis, User Interface Analysis, Document Analysis - Planning and Preparing for elicitation on your project - Performing elicitation activities - Follow up after elicitation	Offline	LO2, LO3	I,T, U	1. Use Case Template.docx 2. Chapter 7 + 8 +Slides		Finish Use Case document	
14	Requirements elicitation (cnt) - Classifying customer input - Some cautions about elicitation - Assumed and implied requirements - Finding missing requirements	Offline	LO2, LO3	I,T, U	1. Use Case Template.docx 2. Chapter 7 + 8 +Slides		Finish Use Case document	
15	Understanding user requirements - Use cases and user stories	Offline	LO5, LO6	T,U	1. Use Case Template.docx 2. Chapter 7 + 8 +Slides		Finish Use Case document	
16	Understanding user requirements (cnt) - The use case approach - Benefits of usage-centric requirements	Offline	LO5, LO6	T,U	1. Use Case Template.docx 2. Chapter 7 + 8 +Slides		Finish Use Case document	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
17	Playing by the rules - A business rules taxonomy - Documenting business rules - Discovering business rules - Business rules and requirements - Tying everything together	Offline	LO4, LO5	T,U	1.Software Requirements Specification Template.docx 2. Chapter 9 + 10 +Slides		Preparing the structure of SRS document (based on student's assignment)	
18	Progress test 1 and Review	Offline	LO1,LO2,LO3,LO4,LO5,LO6		LSM questions		Quiz test	
19	Documenting the requirements - The software requirements specification	Offline	LO7	T,U	1.Software Requirements Specification Template.docx		Preparing the structure of SRS document (based on student's assignment)	
20	Documenting the requirements (cnt) - A software requirements specification template - Requirements specification on agile projects	Offline	LO7	T,U	2. Chapter 9 + 10 +Slides		Preparing the structure of SRS document (based on student's assignment)	
21	Writing excellent requirements - Characteristics of excellent requirements - Guidelines for writing requirements - Sample requirements, before and after	Offline	LO5	T,U	Chapter 11 + Slides		Write SRS document based on guidelines	
22	A picture is worth 1024 words - Data flow diagram; Swimlane diagram; State-transition diagram and state table ; Dialog map ;	Offline	LO5	T,U	Chapter 12 + Slides		Write SRS document based on guidelines	
23	A picture is worth 1024 words (con't) - Decision tables and decision trees;	Offline	LO5	T,U	Chapter 12 + Slides		Write SRS document based on guidelines	
24	A picture is worth 1024 words (con't) - Event-response tables ; Modeling on agile projects	Offline	LO5	T,U	Chapter 12 + Slides		Write SRS document based on guidelines	
25	Specifying data requirements - Modeling data relationships - The data dictionary	Offline	LO5	T,U	1.Guidance for Data Dictionaries.docx 2.Chapter 13 +Slides		Finish Data Dictionaries of Assignment	
26	Specifying data requirements (cnt) - Data analysis - Specifying reports	Offline	LO5	T,U	1.Guidance for Data Dictionaries.docx 2.Chapter 13 +Slides		Finish Data Dictionaries of Assignment	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
27	Beyond functionality - Software quality attributes - Exploring quality attributes - Defining quality requirements - Specifying quality requirements with Planguage	Offline	LO5	T,U	Chapter 14 + Slides		Finish Data Dictionaries of Assignment	
28	Beyond functionality (cnt) - Quality attribute trade-offs - Implementing quality attribute requirements - Constraint - Handling quality attributes on agile projects	Offline	LO5	T,U	Chapter 14 + Slides		Finish Data Dictionaries of Assignment	
29	Risk reduction through prototyping - Prototyping: What and why - Mock-ups and proofs of concept - Throwaway and evolutionary prototypes - Paper and electronic prototypes	Offline	LO8	T,U	Chapter 15 + Slides		Finish Requirements Prioritization worksheet	
30	Risk reduction through prototyping (cnt) - Working with prototypes - Prototype evaluation - Risks of prototyping - Prototyping success factors	Offline	LO8	T,U	Chapter 15 + Slides		Finish Requirements Prioritization worksheet	
31	First things first: Setting requirement priorities - Why prioritize requirements? - Some prioritization pragmatics - Games people play with priorities	Offline	LO2	T,U	1. Requirements Prioritization Spreadsheet.docx 2. Chapter 16 + Slides		Finish Requirements Prioritization worksheet	
32	First things first: Setting requirement priorities (cnt) - Some prioritization techniques - Prioritization based on value, cost, and risk	Offline	LO2	T,U	1. Requirements Prioritization Spreadsheet.docx 2. Chapter 16 + Slides		Finish Requirements Prioritization worksheet	
33	Validating the requirements - Validation and verification - Reviewing requirements - Prototyping requirements	Offline	LO3	T	1. Requirements Review Checklist.docx 2. Chapter 17 +Slides		Check progress of Assignment's requirements	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
34	Validating the requirements - Testing the requirements - Validating requirements with acceptance criteria	Offline	LO3	T	1. Requirements Review Checklist.docx 2. Chapter 17 +Slides		Check progress of Assignment's requirements	
35	Exercises (can be interspersed with the lecture)	Offline	LO3, LO5	I,T,U	1. Chapter 18 + 19 2. Requirements Estimation Tool.xlsx +Slides		Do requirements estimation	
36	Discuss and/or present on Chapter 6	Offline	LO3, LO5	I,T,U	1. Chapter 18 + 19 2. Requirements Estimation Tool.xlsx +Slides		Do requirements estimation	
37	Requirements reuse - Why reuse requirements? - Dimensions of requirements reuse - Types of requirements information to reuse - Common reuse scenarios - Requirement patterns - Tools to facilitate reuse - Making requirements reusable - Requirements reuse barriers and success factors	Offline			1. Chapter 18 2. Requirements Estimation Tool.xlsx +Slides			
38	Beyond requirements development - Estimating requirements effort - From requirements to project plans,designs and code, tests, success Review the first half of assignment and evaluation	Offline	LO2,LO3,LO5,LO7,LO8		1. Chapter 189 Assignment documents		Oral test	
39	Review the first half of assignment and evaluation	Offline	LO2,LO3,LO5,LO7,LO8		Assignment documents Assignment document templates		Oral test	
40	Progress test 2 and Review - Review LAB	Offline	LO2,LO3,LO5,LO7,LO8		LSM questions		Quiz test	
41	Agile projects - Limitations of the waterfall - The agile development approach - Essential aspects of an agile approach to requirements - Adapting requirements practices to agile projects - Transitioning to agile: Now what?	Offline	LO4	I,T,U	Chapter 20+Slides			

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
42	Enhancement and replacement projects - Expected challenges - Requirements techniques when there is an existing system - Prioritizing by using business objectives - When old requirements don't exist - Encouraging new system adoption	Offline	LO4	I,T,U	Chapter 21 +Slides			
43	Packaged solution projects - Requirements for selecting packaged solutions - Requirements for implementing packaged solutions - Common challenges with packaged solution	Offline	LO4	I,T,U	1. Chapter 22 2. Request for Proposal Template.docx 3. Slides		Read and understand the proposal template (what are contents and the purpose of these ones)	
44	Outsourced projects - Appropriate levels of requirements detail - Acquirer-supplier interactions - Change management - Acceptance criteria	Offline	LO4	I,T,U	1. Chapter 23 2. Request for Proposal Template.docx 3. Slides		Read and understand the proposal template (what are contents and the purpose of these ones)	
45	Group discussion Summary and Exercises	Offline			1. Chapter 24 + 25 + Slides		Read and understand the proposal template (what are contents and the purpose of these ones)	
46	Group discussion (cnt) Summary and Exercises (cnt)	Offline			1. Chapter 24 + 25 + Slides		Read and understand the proposal template (what are contents and the purpose of these ones)	
47	Assignment supporting	Offline			1. Chapter 26 + Slides		Checking SRS document	
48	Assignment supporting (cnt)	Offline			1. Chapter 26 + Slides		Checking SRS document	
49	Requirements management practices - Requirements management process - The requirements baseline - Requirements version control - Requirement attributes	Offline	LO4	I,T,U	1. Chapter 25 + 26 + 27 2. a. CCB Charter Template.docx b. Change Control Process.docx c. Impact Analysis Checklist.docx 3. Slides		Do assignment - based on guidelines	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
50	Requirements management practices (cnt) - Tracking requirements status - Resolving requirements issues - Measuring requirements effort - Managing requirements on agile projects	Offline	LO4	I,T,U	1. Chapter 25 + 26 + 27 2. a. CCB Charter Template.docx b. Change Control Process.docx c. Impact Analysis Checklist.docx 3. Slides		Do assignment - based on guidelines	
51	Change happens - Change control policy - Basic concepts of the change control process - A change control process description - The change control board	Offline	LO9	I,T,U	1. Chapter 25 + 26 + 27 2. a. CCB Charter Template.docx b. Change Control Process.docx c. Impact Analysis Checklist.docx 3. Slides		Do assignment - based on guidelines	
52	Change happens (cnt) - Change control tools - Measuring change activity - Change impact analysis - Change management on agile projects	Offline	LO9	I,T,U	1. Chapter 25 + 26 + 27 2. a. CCB Charter Template.docx b. Change Control Process.docx c. Impact Analysis Checklist.docx 3. Slides		Do assignment - based on guidelines	
53	Links in the requirements chain - Tracing requirements - Motivations for tracing requirements - The requirements traceability matrix	Offline	LO9	I,T,U	1. Chapter 25 + 26 + 27 2. a. CCB Charter Template.docx b. Change Control Process.docx c. Impact Analysis Checklist.docx 3. Slides		Do assignment - based on guidelines	
54	Links in the requirements chain (cnt) - Tools for requirements tracing - A requirements tracing procedure	Offline	LO9	I,T,U	1. Chapter 25 + 26 + 27 2. a. CCB Charter Template.docx b. Change Control Process.docx c. Impact Analysis Checklist.docx 3. Slides		Do assignment - based on guidelines	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
55	Tools for requirements engineering (self-study) Improving your requirements processes - How requirements relate to other project processes - Requirements and various stakeholder groups - Gaining commitment to change - Fundamentals of software process improvement - Root cause analysis - The process improvement cycle - Requirements engineering process assets - Creating a requirements process improvement road map	Offline	LO8, LO9	ITU	1. Chapter 31 2. Action Plan Template.docx 3. Risk Management Plan Template.docx 4. Slides		Finish assignment's documents	
56	Software requirements and risk management - Fundamentals of software risk management - Requirements-related risks - Risk management is your friend	Offline	LO8, LO9	ITU	1. Chapter 32 2. Action Plan Template.docx 3. Risk Management Plan Template.docx 4. Slides		Finish assignment 's documents	
57	Progress test 3	Offline	LO4,LO8,LO9		LSM questions		Quiz test	
58	Assignment evaluation	Offline	All LOs		Assignment documents		Oral test	
59	Assignment evaluation	Offline	All LOs		Assignment documents		Oral test	
60	Assignment evaluation	Offline	All LOs		Assignment documents		Oral test	

0 Constructive question(s)

6 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	1	20.0%	>0	Option 1: 26 slots ; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8	Option 1: Teacher raise projects, students practice step by step to complete their assignments. Student could base on study guide COS_Document_Example to do assignments. Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: Depending on questions in assignment Option 2 (For Constructivism Approach only): Follow lecturer's proposal	All subjects in syllabus	in class, by teacher	

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
2	LAB	on-going	1	10.0%	>0	Option 1: 26slots/each Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO5	1. Model 1 Context diagram based on the topic of assignment. 2. Model 1 Swimlane diagam based on the topic of assignment. 3. Model 1 State diagam based on the topic of assignment.	Option 1: Depending on questions in assignment Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Chapter 5, Chapter 12	In class, by teacher	The o of LA draw Conte diagra Swim diagra and 1 diagra These diagra will b drawn indivi and w prese by ea stude week After prese LAB, stude put th diagra on his her Assig docu
3	Progress Test	on-going	3	20.0%	>0	Option 1: 30'/each; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8, LO9	Option 1: Multiple choices Marked by Computer or a suitable format; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 30/each; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Test 1: Chapter 1→ Chapter 9 Test 2: Chapter 10 →20 Test 3: Chapter 21 →32	in class, by LMS system	
4	Final exam	Final exam	2	50.0%	4	145 minutes	All CLOs			All subjects in syllabus	By exam board Manual evaluating	Each of the exam be >= & Fina Resul >=5/1
5	Practical Exam	Final exam	1	25.0%	4	85'	LO1, LO2, LO3, LO4, LO5, LO7, LO8, LO9			All subjects in syllabus	LSM system	Custo from Assig or pro that stude have learne
6	Theory Exam	Final exam	1	25.0%	4	60'/each	LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8, LO9	Multiple choicesMarked by Computer	60	All subjects in syllabus	LSM system	